

ENTERPRISE AZURE FRONT DOOR PREMIUM + API MANAGEMENT (APIM)

ZERO-TRUST ARCHITECTURE & COMPLETE STUDY WORKBOOK

[A4 PRINT EDITION — 30-PAGE MASTER ARCHITECTURE GUIDE]

WORKBOOK OWNERSHIP & PROJECT METADATA

Lead Cloud Architect:			
Target Enterprise Domain:	api.yourcompany.com		
Subscription ID:		Environment:	☐ Dev ☐ Staging ☐ Prod
APIM Gateway Name:		APIM SKU:	☐ Dev ☐ Standardv2 ☐ Premium
Front Door Profile:		WAF Policy:	
Workbook Start Date:		Target Sign-Off:	

ARCHITECTURE TOPOLOGY OVERVIEW & CORE OBJECTIVES

This master guide establishes an enterprise-grade perimeter security and global traffic routing topology. By positioning Azure Front Door (AFD) Premium in front of Azure API Management (APIM), the architecture guarantees:

- Global Traffic Acceleration:** Split-TCP Anycast edge routing for minimal latency across points of presence.
- Layer 7 Edge Defense:** Web Application Firewall (WAF) with OWASP CRS Top 10, Bot Manager, and custom rate limiting.
- Zero-Trust Perimeter Lockdown:** Direct access elimination via X-Azure-FDID header checks, NSG Service Tags, and VNet Private Link.
- Operational Observability:** KQL security queries, health probing (/status-0123456789), and automated TLS lifecycle management.

ARCHITECTURAL PILLARS AT A GLANCE

[GLOBAL EDGE]	[PERIMETER WAF]	[ZERO-TRUST GATEWAY]
- Anycast POP Edge Nodes - Auto-Rotating Managed TLS - Custom Domain CNAME DNS	- Managed OWASP DRS Top 10 - Bot Protection Rulesets - Custom Rate-Limiting Rules	- X-Azure-FDID Header Check - Subnet NSG Service Tags - VNet Private Link Service

QUICK-START INSTRUCTIONS FOR PRINT & STUDY

- SCAN QR CODES:** Point your mobile camera at any QR code in the guide to launch video tutorials or Microsoft documentation.
- USE RULED PAGES:** Turn to the 3 dedicated 8mm notebook sheets following each phase to draft your notes, parameters, and sketches.
- TICK PROGRESS:** Update Page 2 (Master Progress Tracker) after completing each video, document, lab, and security test script.

MASTER LEARNING PROGRESS TRACKER & IMPLEMENTATION DASHBOARD

Tick off each milestone as you progress through video modules, documentation, notes, and lab deployments



END-TO-END IMPLEMENTATION PROGRESS CHECKLIST (TICK BOXES)

Phase Title	Video []	Docs []	Notes []	Lab Deployed []	Tested []	Self-Mastery
Phase 1: APIM Baseline Setup	☐	☐	☐	☐	☐	★ ★ ★ ★ ★
Phase 2: Front Door & Custom SSL	☐	☐	☐	☐	☐	★ ★ ★ ★ ★
Phase 3: WAF OWASP & Rate Limits	☐	☐	☐	☐	☐	★ ★ ★ ★ ★
Phase 4: Perimeter Lockdown & Private Link	☐	☐	☐	☐	☐	★ ★ ★ ★ ★
Phase 5: Pen Testing & Bypass Check	☐	☐	☐	☐	☐	★ ★ ★ ★ ★
Phase 6: Operations Handover & KQL	☐	☐	☐	☐	☐	★ ★ ★ ★ ★



STUDY & LAB HOURS LOG BOOK

Date	Phase	Hours Spent	Topics Studied & Lab Activities Completed
__/__/2026	Phase ____	__ hrs	_____
__/__/2026	Phase ____	__ hrs	_____
__/__/2026	Phase ____	__ hrs	_____
__/__/2026	Phase ____	__ hrs	_____
__/__/2026	Phase ____	__ hrs	_____



KEY MILESTONE BADGES & GOAL COMPLETION SIGN-OFF



Milestone 1

APIM + Front Door Live

☐ Verified



Milestone 2

WAF OWASP Prevention

☐ Verified



Milestone 3

Zero-Trust Lockdown

☐ Verified

PHASE 1: AZURE API MANAGEMENT (APIM) CORE SETUP

Gateway Provisioning, Backend Integration, Products & Policy Hierarchy

PHASE 1: AZURE API MANAGEMENT (APIM) BASELINE SETUP

Duration: 1-2 Days

- **Service Provisioning:** Deploy APIM (Developer, Standard v2, or Premium SKU depending on VNet requirements). Configure global settings, custom domain certificates, and managed identity.
- **Backend Integration & Products:** Import backend REST/GraphQL microservices into APIM. Organize APIs into logical Products (e.g., Public-Starter, Enterprise-Partner).
- **Policy Configurations:** Apply product/global policies including rate limiting (`rate-limit-by-key`), JWT validation (`jwt-validate`), and CORS.
- **Built-in Health Endpoint:** APIM provides a built-in health probe endpoint at `/status-0123456789` (returns `200 OK` when healthy).

 **Video Tutorial Module:**
Azure API Management Deep Dive (John Savill):
<https://www.youtube.com/watch?v=PXtFq5wmGt0>



PHASE 1 - APIM BASELINE SETUP: VIDEO KEY TAKEAWAYS & ARCHITECTURE NOTES

Date Studied: _____

Study Duration: _____ mins

Mastery Level: ★ ★ ★ ★ ★

🎥 VIDEO TUTORIAL CONCEPTS & ARCHITECTURAL TAKEAWAYS (8MM RULED NOTEBOOK LINES)

This image shows a single page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

PHASE 1 - APIM BASELINE SETUP: CONFIGURATION SPECS & RESOURCE PARAMETERS

Azure Resource Group: _____

Subscription ID: _____

⚙️ DETAILED RESOURCE PARAMETERS, POLICIES & CLI COMMANDS (8MM RULED)

This image shows a single page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There is no text or other markings on the page.

PHASE 1 - APIM BASELINE SETUP: COMPONENT FLOW CANVAS & VERIFICATION CHECKLIST

Deployment Status: ☐ Pending ☐ In-Progress ☐ Complete Sign-Off Lead: _____

 PHASE 1 ARCHITECTURE DIAGRAM & FLOW CANVAS

[Freehand Sketch Canvas for Phase 1 Traffic Flow & Integration Topology]

 PHASE 1 EXECUTION & VERIFICATION CHECKLIST

- ☐ 1. Watched recommended video tutorial modules and scanned resource QR codes.
- ☐ 2. Read official Microsoft Learn architectural documentation.
- ☐ 3. Provisioned required Azure resources and configured parameters.
- ☐ 4. Executed validation test scripts and verified expected HTTP response codes.

PHASE 2: AZURE FRONT DOOR (AFD) & CUSTOM DOMAIN SETUP

Profiles, Endpoints, DNS Validation, Managed TLS & Health Probing

PHASE 2: AZURE FRONT DOOR (AFD) & CUSTOM DOMAIN PROVISIONING

Duration: 1-2 Days

- **Profile & Endpoint Creation:** Provision Azure Front Door Premium (required for Private Link and advanced WAF capabilities). Create AFD Endpoint (e.g., ep-api-prod.azurefd.net).
- **Custom Domain & TLS/SSL:** Add Custom Domain (e.g., api.yourcompany.com). Create DNS validation records (CNAME / TXT). Enable Azure-Managed Certificate or bind custom Key Vault certificate.
- **Origin Group & Health Probes:** Point Origin to APIM hostname (apim-prod.azure-api.net). Configure Health Probe Path to /status-0123456789 on HTTPS Port 443 with 30s probe intervals.

 Video Tutorial Module:

Azure Front Door Tutorial: Setup, Configuration & Best Practices:
<https://www.youtube.com/watch?v=cB-jlc6aWXA>



PHASE 2 - FRONT DOOR & CUSTOM DOMAIN: VIDEO KEY TAKEAWAYS & ARCHITECTURE NOTES

Date Studied: _____

Study Duration: _____ mins

Mastery Level: ★ ★ ★ ★ ★

🎥 VIDEO TUTORIAL CONCEPTS & ARCHITECTURAL TAKEAWAYS (8MM RULED NOTEBOOK LINES)

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PHASE 2 - FRONT DOOR & CUSTOM DOMAIN: CONFIGURATION SPECS & RESOURCE PARAMETERS

Azure Resource Group: _____

Subscription ID: _____

DETAILED RESOURCE PARAMETERS, POLICIES & CLI COMMANDS (8MM RULED)

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PHASE 2 - FRONT DOOR & CUSTOM DOMAIN: COMPONENT FLOW CANVAS & VERIFICATION CHECKLIST

Deployment Status: ☐ Pending ☐ In-Progress ☐ Complete Sign-Off Lead: _____

 PHASE 2 ARCHITECTURE DIAGRAM & FLOW CANVAS

[Freehand Sketch Canvas for Phase 2 Traffic Flow & Integration Topology]

 PHASE 2 EXECUTION & VERIFICATION CHECKLIST

- ☐ 1. Watched recommended video tutorial modules and scanned resource QR codes.
- ☐ 2. Read official Microsoft Learn architectural documentation.
- ☐ 3. Provisioned required Azure resources and configured parameters.
- ☐ 4. Executed validation test scripts and verified expected HTTP response codes.

PHASE 3: ENTERPRISE WEB APPLICATION FIREWALL (WAF) SECURITY

Managed OWASP Core Rulesets, Bot Protection & Custom Rate-Limiting

PHASE 3: ENTERPRISE WAF POLICY & CUSTOM SECURITY RULES

Duration: 2 Days

- **WAF Mode Assignment:** Attach a WAF Policy to Front Door Security Policies in **Prevention Mode**.
- **Managed Rulesets:** Enable Default Rule Set (DRS / OWASP Top 10) to guard against SQLi, XSS, and RCE attacks. Enable Bot Manager Ruleset to block malicious scrapers.
- **Custom Rules:** Configure IP Restrictions (Allow/Deny specific CIDR blocks or Geo-locations) and Rate Limiting (e.g., threshold of max 500 requests per 5 minutes per client IP).

Video Tutorial Module:

Azure Front Door WAF Tutorial: OWASP & Rate Limiting:

<https://www.youtube.com/watch?v=d76gE0-D-D4>



PHASE 3 - WAF OWASP & CUSTOM RULES: VIDEO KEY TAKEAWAYS & ARCHITECTURE NOTES

Date Studied: _____

Study Duration: _____ mins

Mastery Level: ★ ★ ★ ★ ★

🎥 VIDEO TUTORIAL CONCEPTS & ARCHITECTURAL TAKEAWAYS (8MM RULED NOTEBOOK LINES)

[illegible]

PHASE 3 - WAF OWASP & CUSTOM RULES: CONFIGURATION SPECS & RESOURCE PARAMETERS

Azure Resource Group: _____

Subscription ID: _____

⚙️ DETAILED RESOURCE PARAMETERS, POLICIES & CLI COMMANDS (8MM RULED)

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PHASE 3 - WAF OWASP & CUSTOM RULES: COMPONENT FLOW CANVAS & VERIFICATION CHECKLIST

Deployment Status: ☐ Pending ☐ In-Progress ☐ Complete Sign-Off Lead: _____

 PHASE 3 ARCHITECTURE DIAGRAM & FLOW CANVAS

[Freehand Sketch Canvas for Phase 3 Traffic Flow & Integration Topology]

 PHASE 3 EXECUTION & VERIFICATION CHECKLIST

- ☐ 1. Watched recommended video tutorial modules and scanned resource QR codes.
- ☐ 2. Read official Microsoft Learn architectural documentation.
- ☐ 3. Provisioned required Azure resources and configured parameters.
- ☐ 4. Executed validation test scripts and verified expected HTTP response codes.

PHASE 4: AFD TO APIM INTEGRATION & ZERO-TRUST PERIMETER LOCKDOWN

Enforcing X-Azure-FDID Header Validation, NSG Service Tag Filtering, and Private Link

PHASE 4: DETAILED INTEGRATION & PERIMETER LOCKDOWN MANUAL

Duration: 1-2 Days

1. Header Lockdown via APIM Global Inbound Policy

- Retrieve Front Door Header ID (X-Azure-FDID) from Front Door Overview.
- In APIM, navigate to **All APIs -> Inbound processing -> Policy editor** and apply:

```
<policies>
  <inbound>
    <check-header name="X-Azure-FDID" failed-check-httpcode="403" failed-check-error-message="Forbidden:
Direct access strictly prohibited" ignore-case="false">
      <value>YOUR-FRONT-DOOR-ID-HERE</value>
    </check-header>
  </inbound>
</policies>
```

2. Network Security Group (NSG) Inbound Lockdown

- Attach NSG to APIM Subnet: Source: ServiceTag / AzureFrontDoor.Backend, Protocol: TCP, Port: 443, Action: Allow. Deny all other internet traffic.

3. Zero-Trust Private Link Setup (Front Door Premium)

- Deploy APIM in Internal VNet mode with Private Link Service targeting internal load balancer. Enable Private Link in Front Door Origin settings.

 **Official Microsoft Guides:**
<https://learn.microsoft.com/en-us/azure/api-management/front-door-api-management>
<https://learn.microsoft.com/en-us/azure/frontdoor/standard-premium/how-to-enable-private-link-apim>



Mastery Level: ★ ★ ★ ★ ★

[illegible]

PHASE 4 - ZERO-TRUST PERIMETER LOCKDOWN: CONFIGURATION SPECS & RESOURCE PARAMETERS

Azure Resource Group: _____

Subscription ID: _____

⚙️ DETAILED RESOURCE PARAMETERS, POLICIES & CLI COMMANDS (8MM RULED)

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PHASE 4 - ZERO-TRUST PERIMETER LOCKDOWN: COMPONENT FLOW CANVAS & VERIFICATION CHECKLIST

Deployment Status: ☐ Pending ☐ In-Progress ☐ Complete

Sign-Off Lead: _____

PHASE 4 ARCHITECTURE DIAGRAM & FLOW CANVAS

[Freehand Sketch Canvas for Phase 4 Traffic Flow & Integration Topology]

PHASE 4 EXECUTION & VERIFICATION CHECKLIST

- ☐ 1. Watched recommended video tutorial modules and scanned resource QR codes.
- ☐ 2. Read official Microsoft Learn architectural documentation.
- ☐ 3. Provisioned required Azure resources and configured parameters.
- ☐ 4. Executed validation test scripts and verified expected HTTP response codes.

PHASE 5: SECURITY TESTING & VALIDATION RUNBOOK

Executing Direct Bypass, Custom Domain Routing, WAF SQLi, and Rate Limit Verification

PHASE 5: COMPREHENSIVE SECURITY VALIDATION MANUAL

Duration: 1 Day

Execution Commands & Expected Outcomes:

- **Direct APIM Bypass Check:** `curl -i -X GET https://apim-prod.azure-api.net/status-0123456789` → 403 Forbidden
- **Front Door Custom Domain Routing Check:** `curl -i -X GET https://api.yourcompany.com/v1/health` → 200 OK
- **WAF OWASP Injection Test (SQLi):** `curl -i "https://api.yourcompany.com/v1/users?id=1' OR '1'='1"` → 403 Forbidden
- **WAF Rate Limit Threshold Test:** Execute burst curl loop exceeding threshold → 429 Too Many Requests

📖 **Official Best Practices:**
<https://learn.microsoft.com/en-us/azure/frontdoor/best-practices>



PHASE 5 - SECURITY PENETRATION TESTING: VIDEO KEY TAKEAWAYS & ARCHITECTURE NOTES

Date Studied: _____

Study Duration: _____ mins

Mastery Level: ★ ★ ★ ★ ★

🎥 VIDEO TUTORIAL CONCEPTS & ARCHITECTURAL TAKEAWAYS (8MM RULED NOTEBOOK LINES)

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PHASE 5 - SECURITY PENETRATION TESTING: CONFIGURATION SPECS & RESOURCE PARAMETERS

Azure Resource Group: _____

Subscription ID: _____

DETAILED RESOURCE PARAMETERS, POLICIES & CLI COMMANDS (8MM RULED)

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PHASE 5 - SECURITY PENETRATION TESTING: COMPONENT FLOW CANVAS & VERIFICATION CHECKLIST

Deployment Status: ☐ Pending ☐ In-Progress ☐ Complete Sign-Off Lead: _____

 PHASE 5 ARCHITECTURE DIAGRAM & FLOW CANVAS

[Freehand Sketch Canvas for Phase 5 Traffic Flow & Integration Topology]

 PHASE 5 EXECUTION & VERIFICATION CHECKLIST

- ☐ 1. Watched recommended video tutorial modules and scanned resource QR codes.
- ☐ 2. Read official Microsoft Learn architectural documentation.
- ☐ 3. Provisioned required Azure resources and configured parameters.
- ☐ 4. Executed validation test scripts and verified expected HTTP response codes.

PHASE 6: DESIGN REVIEW & OPERATIONS HANDOVER

Design Sign-off, IaC Deliverables, SecOps Runbooks & Log Analytics Queries

PHASE 6: OPERATIONAL HANDOVER & MONITORING SPECIFICATIONS

Duration: 1 Day

- **Design Review:** Review architecture, DNS flow, and TLS certificate renewal strategy with SecOps/Cloud Architecture teams.
- **Handover Package Deliverables:** Infrastructure-as-Code (Terraform / Bicep) templates, Log Analytics KQL queries for security alerts, and Operations Runbook for WAF tuning.

```
AzureDiagnostics
| where ResourceProvider == "MICROSOFT.CDN" and Category == "FrontDoorWebApplicationFirewallLog"
| where action_s == "Block"
| project TimeGenerated, clientIp_s, requestUri_s, ruleName_s, details_matches_s
```

PHASE 6 - DESIGN REVIEW & OPERATIONS HANDOVER: VIDEO KEY TAKEAWAYS & ARCHITECTURE NOTES

Date Studied: _____

Study Duration: _____ mins

Mastery Level: ★ ★ ★ ★ ★

🎥 VIDEO TUTORIAL CONCEPTS & ARCHITECTURAL TAKEAWAYS (8MM RULED NOTEBOOK LINES)

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

PHASE 6 - DESIGN REVIEW & OPERATIONS HANDOVER: CONFIGURATION SPECS & RESOURCE PARAMETERS

Azure Resource Group: _____

Subscription ID: _____

⚙️ DETAILED RESOURCE PARAMETERS, POLICIES & CLI COMMANDS (8MM RULED)

This image shows a single page of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

PHASE 6 - DESIGN REVIEW & OPERATIONS HANDOVER: COMPONENT FLOW CANVAS & VERIFICATION CHECKLIST

Deployment Status: ☐ Pending ☐ In-Progress ☐ Complete Sign-Off Lead: _____

 PHASE 6 ARCHITECTURE DIAGRAM & FLOW CANVAS

[Freehand Sketch Canvas for Phase 6 Traffic Flow & Integration Topology]

 PHASE 6 EXECUTION & VERIFICATION CHECKLIST

- ☐ 1. Watched recommended video tutorial modules and scanned resource QR codes.
- ☐ 2. Read official Microsoft Learn architectural documentation.
- ☐ 3. Provisioned required Azure resources and configured parameters.
- ☐ 4. Executed validation test scripts and verified expected HTTP response codes.

DETAILED ARCHITECTURE LEARNING & STUDY GUIDE

Mastering Azure Front Door Premium, WAF Engine & API Management Security Topology

MODULE 1: APIM ARCHITECTURE & MECHANICS

- **Gateway Components:** Proxies requests, enforces policies, validates JWTs, applies rate limits.
- **Policy Hierarchy:** Sequential execution: *Global > Product > API > Operation* via `<base />`.

 **Recommended Video:** APIM Deep Dive (John Savill):
<https://www.youtube.com/watch?v=PXtFq5wmGt0>



MODULE 2: GLOBAL ANYCAST ROUTING & EDGE TRAFFIC

- **Split-TCP & Anycast:** Terminates TCP at nearest PoP edge node, reducing TLS latency over WAN.

 **Recommended Video:** Azure Front Door Setup:
<https://www.youtube.com/watch?v=cB-jlc6aWXA>



MODULE 3: WEB APPLICATION FIREWALL (WAF) & LAYER 7

- **OWASP CRS & Bot Rulesets:** Blocks SQLi, XSS, and RCE at the edge before hitting APIM.

 **Recommended Video:** Front Door WAF & Rate Limiting:
<https://www.youtube.com/watch?v=d76gE0-D-D4>



ARCHITECTURE & SYSTEM FLOW SKETCHPAD

Topic / Video: _____

Module: _____



NETWORK TOPOLOGY & TRAFFIC FLOW CANVAS (FULL-PAGE DRAWING CANVAS)

[Full-Page Freehand Sketching Area for Edge Anycast -> AFD WAF -> Private Link / Header Check -> APIM -> Microservices]

APIM POLICY XML & INFRASTRUCTURE-AS-CODE SCRATCHPAD

Policy Scope: ☐ Global ☐ Product ☐ API ☐ Operation

laC Framework: ☐ Terraform ☐ Bicep

INBOUND / OUTBOUND POLICY XML DRAFT AREA (LINED RULED CODE CANVAS)

`<!-- Draft your custom policy code (e.g. rate-limit-by-key, jwt-validate, set-header) below: -->`

SECOPS LOG ANALYTICS & KQL QUERY DESIGN SHEET

Log Source: ☐ Front Door WAF Log ☐ APIM Gateway Log

Alert Severity: ☐ Sev 0 ☐ Sev 1 ☐ Sev 2

 KQL QUERY WORKSHOP (LINED RULED QUERY CANVAS)

// Draft custom Log Analytics KQL query for attack detection / anomalous traffic:

TROUBLESHOOTING & INCIDENT RUNBOOK LOG

Resource Title: _____

Date Studied: _____



DEEP STUDY NOTES & DIAGNOSTIC COMMANDS (A4 RULED SPACING - 8MM)